第七届中小学人工智能教育展示活动【教师基本功测试】参考答案

一、选择题

1. D　2.A　3.B　4.C　5.A

二、填空题

(1) pose_features

(2) pose_features

(3) x-min_val

其他合理答案也可以。

三、综合题

1. 参考代码

数据拆分：

```
from BaseDT.dataset import split_tab_dataset
# 指定待拆分的csv数据集
path = "auto-mpg.csv"
# 指定特征数据列、标签列、训练集比重，‘normalize=True’表示进行归一化处理
tx,ty,val_x,val_y = split_tab_dataset(path,data_column=range(0,7),
                                      label_column=7,train_val_ratio=0.8,normalize=False)
```

```
正在划分数据集，比例为 train:val = 0.8:0.2
实际数据量为 train:val = 313:79
划分中......
划分完毕!
训练集保存至auto-mpg_train.csv，验证集保存至auto-mpg_val.csv。
```

模型训练：

```
from BaseML import Regression as reg # 从库文件中导入回归任务模块
model = reg('LinearRegression') # 实例化线性回归模型
model.load_tab_data( './auto-mpg_train.csv') # 载入训练数据
model.train() # 训练模型
model.valid('./auto-mpg_val.csv',metrics='r2') # 载入验证数据并验证
model.save('mymodel.pkl') # 保存模型供应用
```

```
验证r2-score为：83.62752811824078%
Saving model checkpoints...
Saved successfully!
```

模型应用：

```python
import PySimpleGUIWeb as sg
from XEdu.hub import Workflow as wf
import numpy as np

model_path = 'mymodel.pkl'
model = wf(task='baseml',checkpoint=model_path)

def my_inf(data):
    result1 = model.inference(data)
    print(result1)
    return result1

#背景色
sg.theme('LightGreen')
#定义窗口布局
layout = [
    [sg.Input(key='1')],
    [sg.Input(key='2')],
    [sg.Input(key='3')],
    [sg.Input(key='4')],
    [sg.Input(key='5')],
    [sg.Input(key='6')],
    [sg.Input(key='7')],
    [sg.Button('计算', size=(20, 1))],
    [sg.Text('推理结果：',key='res')]
]
```

```python
#窗口设计
window = sg.Window('推理',layout,size=(600, 500))

while True:
    event, values = window.read(timeout=0, timeout_key='timeout')
    if event == '计算':
        data=[int(values['1']),int(values['2']),int(values['3']),int(values['4']),
              float(values['5']),int(values['6']),int(values['7'])]
        #data=np.array(data)
        res = my_inf([data])
        window['res'].update('结果：'+ str(res[0]))
window.close()
```

界面展示：

2. 参考代码

```
# 第一步：提取所有图片的关键点信息，制作关键点数据集
```

```python
import os
import numpy as np
from XEdu.hub import Workflow as wf
body = wf(task='pose_body17')# 实例化pose模型
det = wf(task='det_body')# 实例化detect模型
ds = os.listdir('wqx3_imagenet/training_set')
for d in ds:
    if d[0]=='c':
        continue
    fs = os.listdir(os.path.join('wqx3_imagenet/training_set',d))
    for f in fs:
        path = os.path.join('wqx3_imagenet/training_set',d,f)
        print(path)
        bboxs = det.inference(data=path,thr=0.3)
        if len(bboxs)>0:
            for i in bboxs:
                keypoints = body.inference(data=path,bbox=i)
                array = np.array(keypoints).reshape(1,-1)
                #print(array)
                x_min = array.min()
                x_max = array.max()
                normalized_arr = (array - x_min) / (x_max - x_min) # 做一个归一化
                # 打开文件，如果文件不存在则创建，并追加内容
                with open('train2.csv', 'a') as file:
                    #file.write(d[0])
                    for n in normalized_arr[0]:
                        file.write(str(n)+',')
                    file.write(d[0]+'\n')
                #print(keypoints)
        else:
            pass
```

```
wqx3_imagenet/training_set\2_deerbutt\2_DeerButt_5c2341c28_9148_0168.jpg
wqx3_imagenet/training_set\2_deerbutt\2_DeerButt_607167abb_1543_0054.jpg
wqx3_imagenet/training_set\2_deerbutt\2_DeerButt_607167abb_1543_0060.jpg
wqx3_imagenet/training_set\2_deerbutt\2_DeerButt_607167abb_1543_0066.jpg
wqx3_imagenet/training_set\2_deerbutt\2_DeerButt_670264f89_6274_0110.jpg
```

```python
# 第二步：搭建神经网络训练模型，并保存
```

```python
from BaseNN import nn
model = nn('cls')
train_path = 'train3.csv'
model.load_tab_data(train_path, batch_size=1000)
model.add(layer='linear',size=(34, 64),activation='relu')
model.add(layer='linear',size=(64, 8), activation='relu')
model.add(layer='linear', size=(8, 3), activation='softmax')
model.save_fold = 'checkpoints/'
# 模型训练
model.train(lr=0.01, epochs=1000)
```

```
 {'epoch': 491, 'loss': 0.6184123754501343},
 {'epoch': 491, 'loss': 0.6758401393890381},
 {'epoch': 492, 'loss': 0.6312528848648071},
 {'epoch': 492, 'loss': 0.6252785921096802},
 {'epoch': 493, 'loss': 0.6332786679267883},
 {'epoch': 493, 'loss': 0.6148770451545715},
 {'epoch': 494, 'loss': 0.6341853141784668},
 {'epoch': 494, 'loss': 0.6408733129501343},
 {'epoch': 495, 'loss': 0.6268175840377808},
 {'epoch': 495, 'loss': 0.6376425623893738},
 {'epoch': 496, 'loss': 0.6324204206466675},
 {'epoch': 496, 'loss': 0.6719021201133728},
 {'epoch': 497, 'loss': 0.6486886739730835},
 {'epoch': 497, 'loss': 0.6211971044540405},
 {'epoch': 498, 'loss': 0.6432052254676819},
 {'epoch': 498, 'loss': 0.6368945240974426},
 {'epoch': 499, 'loss': 0.6430871486663818},
 {'epoch': 499, 'loss': 0.6163651347160339},
 ...]
```

验证模型效果，同样要提取验证集关键点

```python
import os
import numpy as np
from XEdu.hub import Workflow as wf
body = wf(task='pose_body17')# 实例化pose模型
det = wf(task='det_body')# 实例化detect模型
ds = os.listdir('wqx3_imagenet/val_set')
for d in ds:
    if d[0]=='c':
        continue
    fs = os.listdir(os.path.join('wqx3_imagenet/val_set',d))
    for f in fs:
        path = os.path.join('wqx3_imagenet/val_set',d,f)
        print(path)
        bboxs = det.inference(data=path,thr=0.3)
        if len(bboxs)>0:
            for i in bboxs:
                keypoints = body.inference(data=path,bbox=i)
                array = np.array(keypoints).reshape(1,-1)
                #print(array)
                x_min = array.min()
                x_max = array.max()
                normalized_arr = (array - x_min) / (x_max - x_min) # 用一样的归一化方法
                # 打开文件，如果文件不存在则创建，并追加内容
                with open('val.csv', 'a') as file:
                    #file.write(d[0])
                    for n in normalized_arr[0]:
                        file.write(str(n)+',')
                    file.write(d[0]+'\n')
                #print(keypoints)
        else:
            pass
```

```
pose_body17任务模型加载成功!
det_body任务模型加载成功!
wqx3_imagenet/val_set\2_deerbutt\2_DeerButt_002bb91db_5023_0090.jpg
wqx3_imagenet/val_set\2_deerbutt\2_DeerButt_0db4081e9_1106_0112.jpg
wqx3_imagenet/val_set\2_deerbutt\2_DeerButt_0e15d0c66_6668_0024.jpg
```

```python
import numpy as np
# 读取验证集
val_path = 'val.csv'
val_x = np.loadtxt(val_path, dtype=float, delimiter=',',skiprows=1,usecols=range(0,34)) # 读取特征列
val_y = np.loadtxt(val_path, dtype=float, delimiter=',',skiprows=1,usecols=34) # 读取预测值列

# 导入库
from BaseNN import nn
# 声明模型
model = nn('reg')
y_pred = model.inference(val_x,checkpoint = 'basenn.pth')  # 对该数据进行预测
```

```python
# 用测试数据查看模型效果
model2 = nn('cls')
test_path = 'val.csv'
test_x = np.loadtxt(test_path, dtype=float, delimiter=',',skiprows=1,usecols=range(0,34))
# 获取最后一列的真实值
test_y = np.loadtxt(test_path, dtype=float, delimiter=',',skiprows=1,usecols=34)

res = model2.inference(test_x, checkpoint="basenn.pth")
model2.print_result(res)

# 定义一个计算分类正确率的函数
def cal_accuracy(y, pred_y):
    res = pred_y.argmax(axis=1)
    tp = np.array(y)==np.array(res)
    acc = np.sum(tp)/ y.shape[0]
    return acc

# 计算分类正确率
print("分类正确率为：",cal_accuracy(test_y, res))
```

```
0}, 105: {'预测值': 1, '置信度': 1.0}, 106: {'预测值': 1, '置信度': 1.0}, 107: {'预测值': 1, '置信度':
'置信度': 0.9999969}, 109: {'预测值': 1, '置信度': 0.9999995}, 110: {'预测值': 1, '置信度': 1.0}, 111:
2: {'预测值': 0, '置信度': 1.0}, 113: {'预测值': 0, '置信度': 0.99999964}, 114: {'预测值': 0, '置信度'
度': 0.9981844}, 116: {'预测值': 2, '置信度': 0.98952043}, 117: {'预测值': 2, '置信度': 0.9985317}, 11
317}, 119: {'预测值': 2, '置信度': 0.9985317}, 120: {'预测值': 2, '置信度': 0.9985317}, 121: {'预测值'
{'预测值': 2, '置信度': 0.9985317}, 123: {'预测值': 2, '置信度': 0.9985317}, 124: {'预测值': 2, '置信度
```

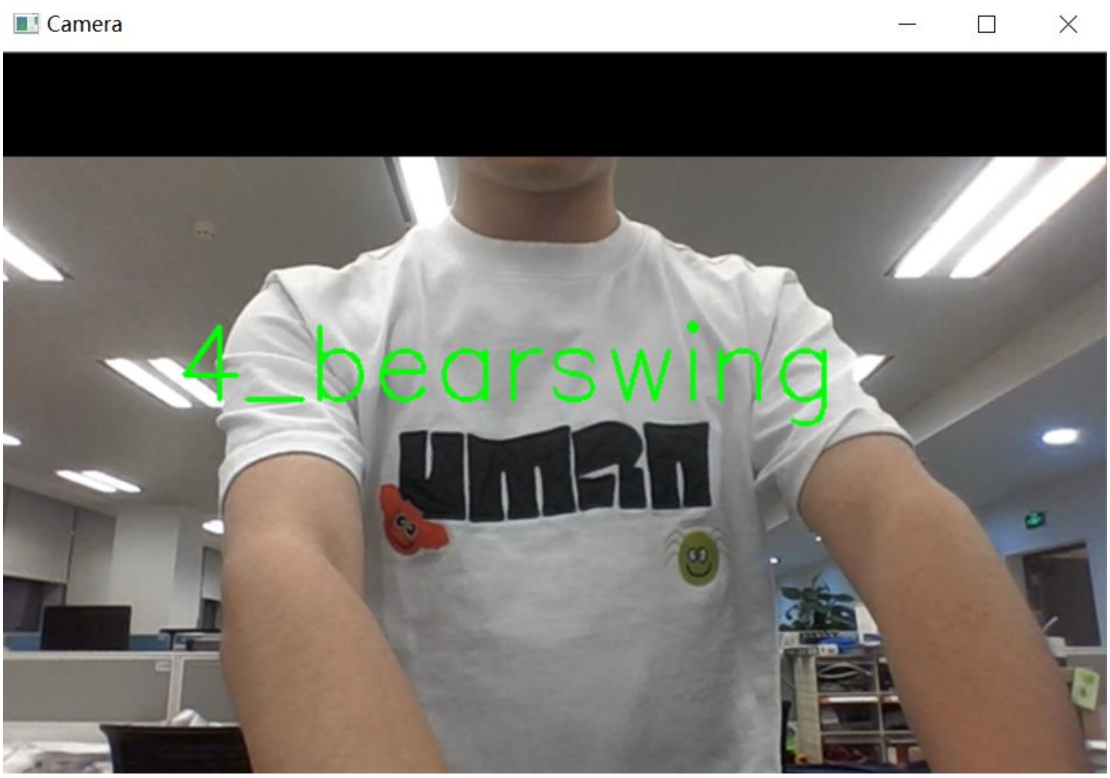
```python
# 模型转换
from BaseNN import nn
model = nn()
model.convert(checkpoint="checkpoints/basenn.pth",out_file="basenn.onnx")
```

```
Successfully exported ONNX model: basenn.onnx
```

```python
# 写一个界面实时预测
import cv2
import numpy as np
from XEdu.hub import Workflow as wf
body = wf(task='pose_body17')# 实例化pose模型
det = wf(task='det_body')# 实例化detect模型
classes = ['2_deerbutt','4_bearswing','6_monkeylift']

model = wf(task='basenn',checkpoint='basenn.onnx')
cap = cv2.VideoCapture(0)

while True:
    ret, frame = cap.read()
    bboxs = det.inference(data=frame,thr=0.3)
    if len(bboxs)>0:
        for i in bboxs:
            keypoints = body.inference(data=frame,bbox=i)
            array = np.array(keypoints).reshape(1,-1)
            x_min = array.min()
            x_max = array.max()
            normalized_arr = (array - x_min) / (x_max - x_min) # 用一样的归一化方法
            res = model.inference(normalized_arr)
            #print(res)
            res = model.format_output(isprint=False)
            print(classes[res[0]['预测值']])

    cv2.putText(frame, classes[res[0]['预测值']], (100, 200), cv2.FONT_HERSHEY_SIMPLEX, 2, (0, 255, 0), 2)
    cv2.imshow("Camera", frame)
    # 按下 'q' 键退出循环
    if cv2.waitKey(1) & 0xFF == ord('q'):
        break

# 关闭摄像头和窗口
cap.release()
cv2.destroyAllWindows()
```

```
6_monkeylift
6_monkeylift
```

界面展示：



附件：

https://share.weiyun.com/s3Vu4B0b